

Giuseppe Nicola Liso

PhD Student · Bayesian Statistics · Gaussian Mixture Models · Machine Learning · Stochastic Processes

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PROFILE

PhD student in Statistics and Computer Science (Statistics Track) at Bocconi University, with fellowship. MSc in Mathematics and Statistics for Life and Social Sciences with highest honours. Research experience at Imperial College London developing Bayesian Gaussian mixture models for multiplex serology. Research topics: Bayesian statistics and Bayesian inference; Gaussian mixture models and mixture distributions; machine learning; stochastic processes.

EDUCATION

August 2026–present

Bocconi University – Milan, Italy

PhD in Statistics and Computer Science, Statistics Track – with Fellowship

- Research topics: Bayesian statistics and Bayesian inference; Gaussian mixture models and mixture distributions; machine learning; stochastic processes.

2024–2026 · Graduated
17 July 2026 · 110/110
cum laude

University of Trento – Trento, Italy

MSc in Mathematics and Statistics for Life and Social Sciences – Mathematics for Data Science (LM-40)

- English-taught programme covering Bayesian statistics, stochastic processes, optimization, machine learning, mathematical modelling, and advanced data analysis.
- Thesis: nonlinear extensions of Bayesian Gaussian mixture models for multiplex serology; model derivation, R/Stan implementation, simulation studies, and application to xSTAR serological data.

2021–2024 · Graduated
19 September 2024 ·
110/110 cum laude

University of Trento – Trento, Italy

BSc in Mathematics (L-35)

- Thesis: Mixture Distributions and the EM Algorithm in the Fight Against Counterfeiting: The Case of Rectified Concentrated Musts.

RESEARCH EXPERIENCE

February–July 2026

Imperial College London – London, United Kingdom

Visiting Research Student, Department of Infectious Disease Epidemiology

- Developed nonlinear extensions of a Bayesian Gaussian mixture framework for cross-reactive antibody responses to dengue, Zika, and chikungunya viruses.
- Derived model-specific mean and covariance structures and implemented Bayesian inference workflows in R and Stan.
- Designed simulation studies assessing convergence, parameter recovery, interval coverage, and sensitivity to priors and initialization.
- Applied models to unpublished xSTAR serological data from multiple African locations; supervised by Dr Ilaria Dorigatti.

TEACHING EXPERIENCE

October 2025–February
2026

University of Trento – Trento, Italy

Mathematics Tutor

- Tutored three university students in Mathematics for Economics, adapting explanations and guided problem solving to individual needs.

PERSONAL PROJECTS

Barcelona Metro Travel Patterns – Tucker3 Decomposition

Applied Tucker3 tensor decomposition to identify spatial clusters, temporal patterns, and interactions in metro passenger flows.

Cigarette Smoking and Airway Epigenetics

Analysed gene expression and DNA methylation changes in small airway epithelium, combining statistical analysis and machine learning across omics data.

Landscapes of Crashes – Topological Analysis of Financial Time Series

Used Topological Data Analysis to study structural changes in financial markets and compared a topological complexity indicator with the VIX over 1987–2016.

TD-Gammon – Temporal-Difference Learning

Developed and analysed a Backgammon agent trained through self-play using temporal-difference learning and neural-network function approximation.

Loan Approval Prediction

Built an end-to-end Python workflow for data cleaning, statistical testing, model comparison, cross-validated Random Forest tuning, calibration, feature importance, and error analysis.

TECHNICAL SKILLS

Statistics and Mathematics: statistics, Bayesian statistics and inference, advanced statistical methods, Gaussian mixture models and mixture distributions, MCMC and HMC/NUTS, model selection, uncertainty quantification, stochastic processes, time-series analysis, numerical analysis, mathematical modelling, optimization methods, simulation studies.

Programming and Computing: Python, R, Stan, SQL, Java, basic C++, Git/GitHub, Jupyter Notebooks, cmdstanr, scientific computing, algorithms and data structures.

Machine Learning and AI: machine learning, deep learning, supervised learning, Random Forests, model validation and calibration, reinforcement learning, neural networks, TensorFlow, PyTorch.

Data Analysis and Visualization: data analysis, data visualization (Matplotlib, ggplot2), relational databases, SQL-based querying, tensor decomposition for data analysis, geometry and topology for data analysis, multi-omic data analysis.

AWARDS AND CERTIFICATIONS

- Galileo Festival of Innovation, Padua (2025): selected student; participation scholarship covering accommodation and board.
- Comau Robotics Certification (2019): 100/100.
- IBM: What is Data Science? (06S7FDUJZVNL); Tools for Data Science (18YGX3IZ4MYL).

LANGUAGES

- Italian: native.
- English: C1.